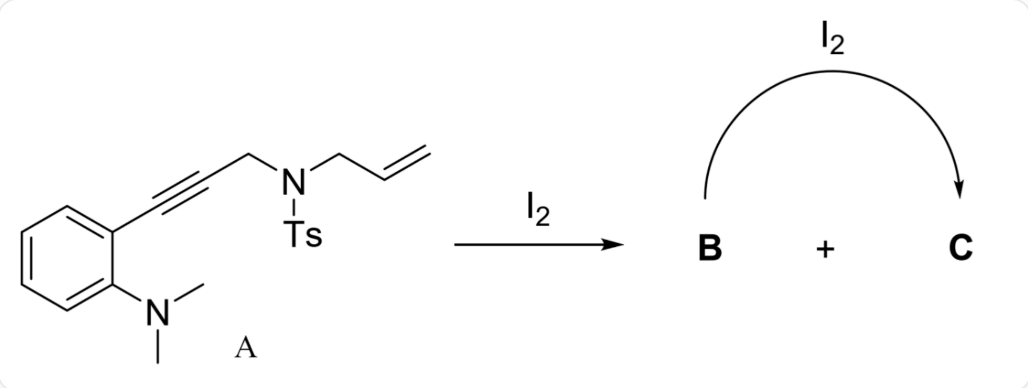
Question

Compound $\mathbf{A}$ can be converted into $\mathbf{B}(C_{20}H_{21}IN_2O_2S)$, $\mathbf{C}(C_{20}H_{21}IN_2O_2S)$ under the action of I_2:



Compound $\mathbf{A}$ is C=CCN(S(C1=CC=C(C)C=C1)(=O)=O)CC#CC2=CC=CC=C2N(C)C, the reaction of $\mathbf{A}$ with I_2 can be converted into $\mathbf{B}$ and $\mathbf{C}$, and the reaction of $\mathbf{B}$ with I_2 can also be converted into $\mathbf{C}$

The following experiments have been carried out:

Entry	Solvent	$\mathbf{A}:I_2$	temp(°C)	Time(h)	Yield($\mathbf{B}$)	Yield($\mathbf{C}$)
1	DCE	1:1	70	4	92	trace
2	DCE	1:2	70	4	0	90

DCE: dichloroethane.

It is known that $\mathbf{B}$ can be converted into $\mathbf{C}$ in DCE under the same conditions with $1\text{eq}\ I_2$.

There are the following statements:

1. Both $\mathbf{B}$ and $\mathbf{C}$ have three rings.

2. The number of "CH_2" groups in $\mathbf{B}$ and $\mathbf{C}$ are the same, and the number of "CH" groups are different.

3. The number of rings in the key intermediate for the formation of $\mathbf{B}$ from $\mathbf{A}$ and the key intermediate for the formation of $\mathbf{C}$ from $\mathbf{B}$ are all 3.

4. If 1 eq of I_2 labeled with a radioisotope is added first to completely convert $\mathbf{A}$ to $\mathbf{B}$, and then 1 eq of unlabeled I_2 is added to convert $\mathbf{B}$ to $\mathbf{C}$, then no radioactivity can be detected in $\mathbf{C}$.

The option that contains all the correct statements is:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1, 2

G. 1, 3

H. 1, 4

I. 2, 3

J. 2, 4

K. 3, 4

L. 1, 2, 3

M. 1, 2, 4

N. 1, 3, 4

O. 2, 3, 4

P. 1, 2, 3, 4

Answer

Correct Answer: **A**

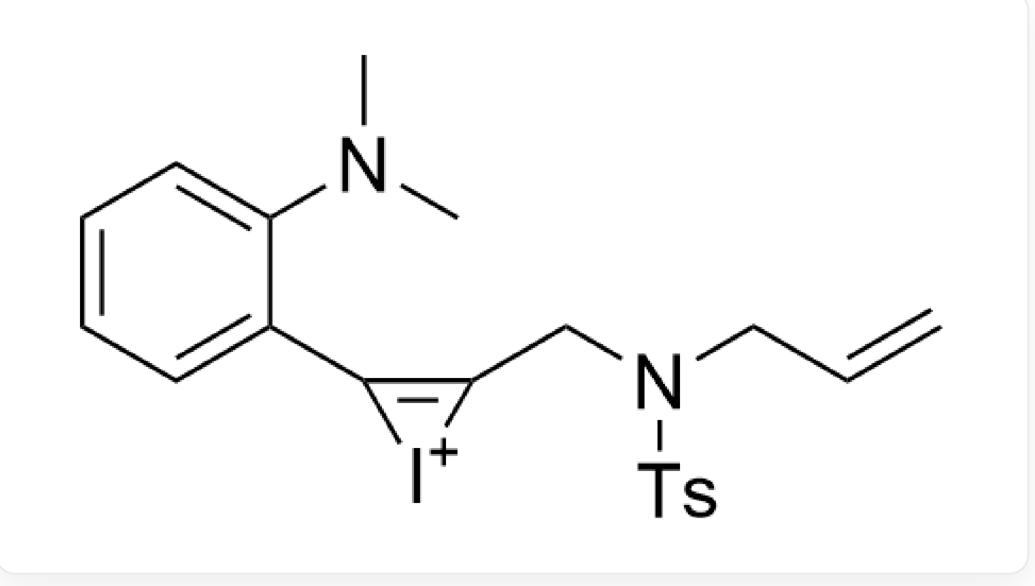
Detailed Explanation

From the reaction ratios given in the table, it can be seen that product **B** is formed upon addition of 1 eq of iodine, and product **C** is formed upon further addition of 1 eq of iodine. Therefore, the reaction mainly consists of two steps, and the formation of product **C** requires more than 1 eq of iodine.

CHECKPOINT 1 PTS

Product **C** is not formed when only 1 eq of iodine is added

This reaction is a typical sequential reaction starting from an iodonium ion. The triple bond in the substrate first forms an iodonium ion intermediate:

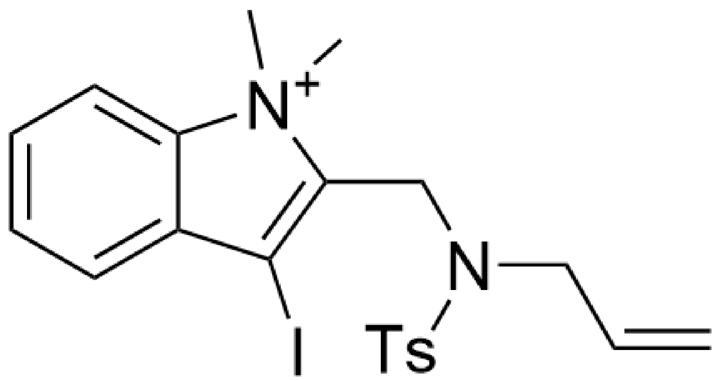


C=CCN(S(C1=CC=C(C)C=C1)(=O)=O)CC2=C([I+]2)C3=CC=CC=C3N(C)C

CHECKPOINT 1 PTS

The first intermediate is: C=CCN(S(C1=CC=C(C)C=C1)(=O)=O)CC2=C([I+]2)C3=CC=CC=C3N(C)C

Subsequently, amine attack occurs, opening the three-membered ring to give the second intermediate:

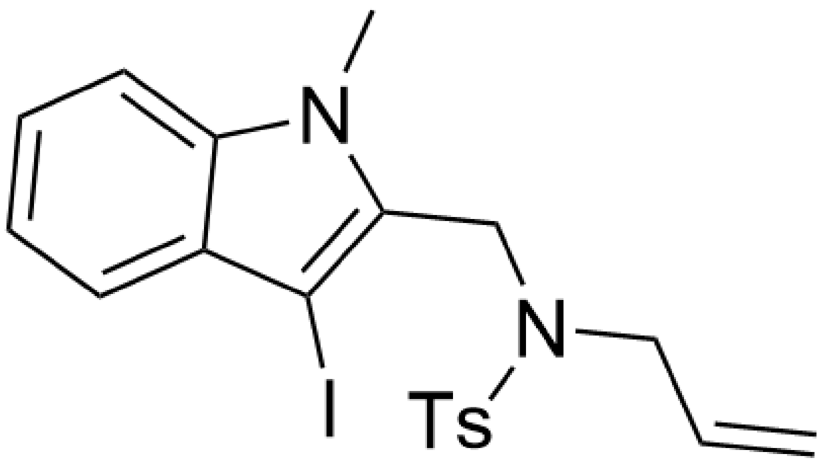


C=CCN(S(C1=CC=C(C)C=C1)(=O)=O)CC2=C(I)C3=CC=CC=C3[N+]2(C)C

CHECKPOINT 1 PTS

The second intermediate is C=CCN(S(C1=CC=C(C)C=C1)(=O)=O)CC2=C(I)C3=CC=CC=C3[N+]2(C)C

Finally, a reaction occurs in which iodide attacks the methyl group, yielding product $\mathbf{B}$:

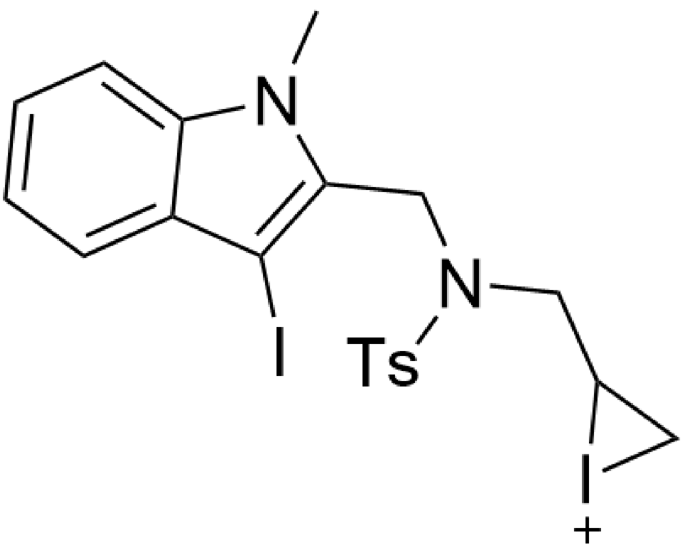


C=CCN(S(C1=CC=C(C)C=C1)(=O)=O)CC2=C(I)C3=CC=CC=C3N2C

CHECKPOINT 1 PTS

B is C=CCN(S(C1=CC=C(C)C=C1)(=O)=O)CC2=C(I)C3=CC=CC=C3N2C

B can undergo a similar reaction as described above once more. First, an iodonium ion is formed at the double bond:

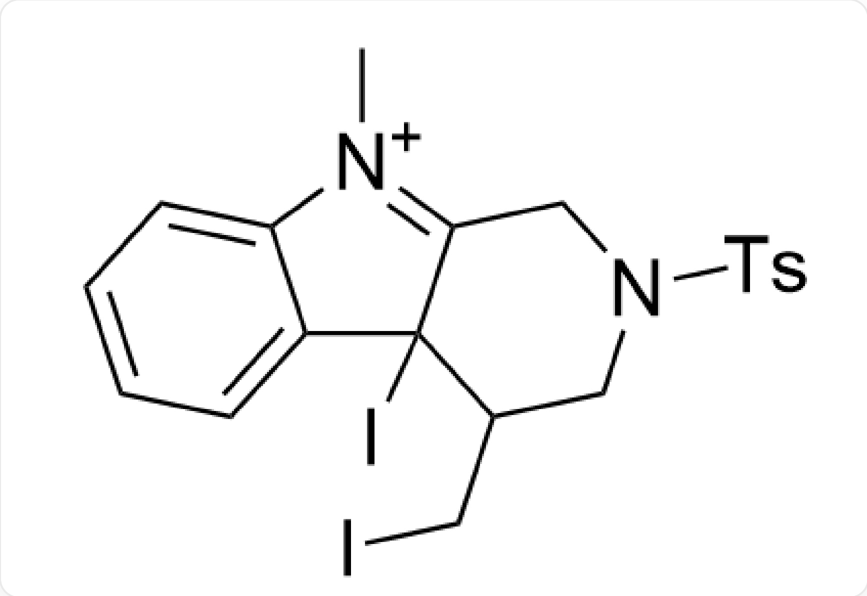


CN1C2=CC=CC=C2C(I)=C1CN(S(C3=CC=C(C)C=C3)(=O)=O)CC4C[I+]4

CHECKPOINT 1 PTS

The third intermediate is CN1C2=CC=CC=C2C(I)=C1CN(S(C3=CC=C(C)C=C3)(=O)=O)CC4C[I+]4

Subsequently, the indole ring's carbon at the 3-position attacks the three-membered ring, generating the next intermediate:

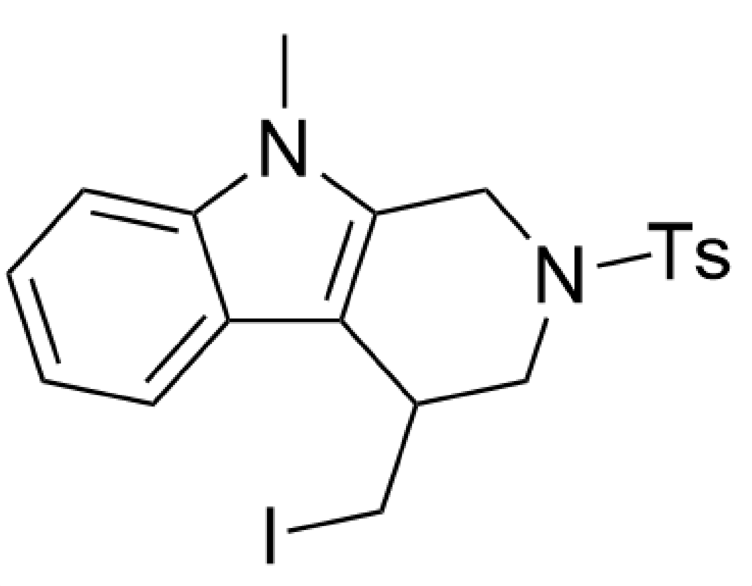


C[N+]1=C2CN(S(C3=CC=C(C)C=C3)(=O)=O)CC(C2(I)C4=CC=CC=C41)CI

CHECKPOINT 1 PTS

The fourth intermediate is C[N+]1=C2CN(S(C3=CC=C(C)C=C3)(=O)=O)CC(C2(I)C4=CC=CC=C41)CI

Finally, an iodide attacks the iodine, eliminating iodine to yield **C**:



CN1C2=CC=CC=C2C3=C1CN(S(C4=CC=C(C)C=C4)(=O)=O)CC3CI

CHECKPOINT 1 PTS

C is CN1C2=CC=CC=C2C3=C1CN(S(C4=CC=C(C)C=C4)(=O)=O)CC3CI

1. **B** has three rings, and **C** has four rings. Incorrect.

2. **B** has three "CH_2" groups and five "CH" groups, and **C** has three "CH_2" groups and five "CH" groups. The numbers are the same. Incorrect.

3. The key intermediates in the formation of **B** from **A** all have 3 rings, and the key intermediates in the formation of **C** from **B** all have 4 rings. Incorrect.

4. In the last step of the formation of **C** from **B**, the initially present radioactive iodine atom will be transferred to iodine,

CHECKPOINT 1 PTS

Radioactive iodine atoms are transferred to iodine when $\mathbf{B}$ is converted to $\mathbf{C}$

The radioactive iodine can then undergo this step again, so radioactivity can also be detected in $\mathbf{C}$. Incorrect.

